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Building Data Lakes on AWS ×

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Reading: AWS Lake Formation Basics

AWS Lake Formation helps you centrally govern, secure, and globally share data for analytics and machine learning. With Lake Formation, you can manage fine-grained access control for your data lake data on Amazon Simple Storage Service (Amazon S3) and its metadata in AWS Glue Data Catalog.

Lake Formation helps you break down data silos and combine different types of structured and unstructured data into a centralized repository. First, identify existing data stores in Amazon S3 or relational and NoSQL databases, and move the data into your data lake. Then crawl, catalog, and prepare the data for analytics. Next, provide your users with secure self-service access to the data through their choice of analytics services.

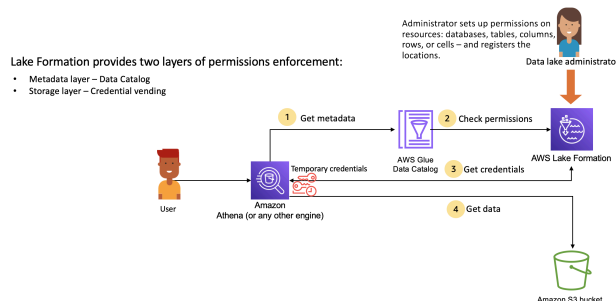
Lake Formation provides its own permissions model that augments the IAM permissions model. Lake Formation permissions model enables fine-grained access to data stored in data lakes through a simple grant or revoke mechanism, much like a relational database management system (RDBMS). Lake Formation permissions are enforced using granular controls at the column, row, and cell-levels across AWS analytics and machine learning services, including Amazon Athena, Amazon QuickSight, Amazon Redshift Spectrum, Amazon EMR, and AWS Glue.

The Lake Formation hybrid access mode for AWS Glue Data Catalog lets you secure and access the cataloged data using both Lake Formation permissions and IAM permissions policies for Amazon S3 and AWS Glue actions. With hybrid access mode, data administrators can onboard Lake Formation permissions selectively and incrementally, focusing on one data lake use case at a time.

Lake Formation also allows you to share data internally and externally across multiple AWS accounts, AWS organizations or directly with IAM principals in another account providing fine-grained access to the AWS Glue Data Catalog metadata and underlying data.

Lake Formation permissions management workflow

Lake Formation integrates with analytical engines to query Amazon S3 data stores and metadata objects that are registered with Lake Formation. The following diagram illustrates how permissions management works in Lake Formation.



Lake Formation permissions management high-level steps

Before Lake Formation can provide access controls for data in your data lake, a data lake administrator or a user with administrative permissions sets up individual Data Catalog table user policies to allow or deny access to Data Catalog tables using Lake Formation permissions.

Then, either the data lake administrator or a user delegated by the administrator grants Lake Formation permissions to users on the Data Catalog databases and tables, and registers the Amazon S3 location of the table with Lake Formation.

1. **Get metadata** – A principal (user) submits a query or an ETL script to an integrated analytical engine such as Amazon Athena, AWS Glue, Amazon EMR, or Amazon Redshift Spectrum. The integrated analytical engine identifies the table that is being requested and sends a request for metadata to the Data Catalog.
2. **Check permissions** – The Data Catalog checks user's permissions with Lake Formation, and if the user is authorized to access the table, returns the metadata that the user is allowed to see to the engine.
3. **Get credentials** – The Data Catalog lets the engine know if the table is managed by Lake Formation or not. If the underlying data is registered with Lake Formation, the analytical engine requests Lake Formation to provide data access by granting temporary access.
4. **Get data** – If the user is authorized to access the table, Lake Formation provides temporary access to the integrated analytical engine. Using the temporary access, the analytical engine fetches the data from Amazon S3, and performs necessary filtering such as column, row, or cell filtering. When the engine finishes running the job, it returns the results back to the user. This process is called credential vending.

If the table is not managed by Lake Formation, the second call from the analytic engine is made directly to Amazon S3. The concerned Amazon S3 bucket policy and IAM user policy are evaluated for data access.

Whenever you use IAM policies, make sure that you follow IAM best practices. For more information, see Security best practices in IAM in the IAM User Guide.

For more information regarding AWS Lake Formation, visit the AWS Lake Formation Developer Guide at <https://docs.aws.amazon.com/lake-formation/latest/dg/what-is-lake-formation.html>

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 **Venkatesh**
Give me a summary

 **Coach**
The content focuses on AWS Lake Formation, a service that helps manage and secure data lakes for analytics and machine learning.

AWS Lake Formation Overview

- Centralizes governance, security, and sharing of data across analytics and machine learning platforms.
- Facilitates the combination of structured and unstructured data into a single repository by moving data from various sources into Amazon S3.

Permissions Management

- Implements a permissions model that allows fine-grained access control similar to relational databases.
- Enforces permissions at the column, row, and cell levels across various AWS services like Amazon Athena and AWS Glue.

Data Access Workflow

- Involves a series of steps where users request data, permissions are checked, and temporary access is granted for data retrieval.
- Supports hybrid access mode, allowing the use of both Lake Formation and IAM permissions for data management.

And if you want to continue exploring this topic, try one of these follow-up questions:

What is the Lake Formation permissions model?

How can you implement fine-grained access control in Lake Formation?

How does Lake Formation integrate with AWS analytics services for data security?

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